Normal Vectors and Quadratic Forms

Linear Algebra · Lecture 10

Last time (Lecture 9): $\operatorname{var}(Az+b) = A\Sigma A'$, $\operatorname{cov}(Az, Bz) = A\Sigma B'$ if $\operatorname{var}(z) = \Sigma$. Earlier results are cited where they are used. $I = I_n$, $e_j = \operatorname{col}_j(I)$.

Facts on random variables (numbers whose values depend on chance), without proof; F3–F11 where first used.

F1. M_z (Definition 1) finite near 0 $\Rightarrow$ at $t = 0$: $\partial M_z/\partial t_i = \mathsf{E}(z_i)$, $\partial^2 M_z/\partial t_j \partial t_i = \mathsf{E}(z_i z_j)$.
F2. $\mathsf{E}(\sum_k c_k u_k + c) = \sum_k c_k \mathsf{E}(u_k) + c$; $\mathsf{E}\, g(u) = \sum_k g(a_k)\, \mathsf{P}(u = a_k)$ if $u \in \{a_1, \dots, a_m\}$.

1. Normal vectors

Definition 1. The **moment generating function** (MGF) of a random vector $z \in \mathbb{R}^n$ (entries random variables; $\mathsf{E}(z) = (\mathsf{E}(z_i))$, $\operatorname{var}(z) = \mathsf{E}\Big((z - \mathsf{E}\, z)(z - \mathsf{E}\, z)'\Big)$, Lecture 9, Definition 1) is $M_z(t) = \mathsf{E} \exp(t'z)$ ($t \in \mathbb{R}^n$, where finite). Let $\mu \in \mathbb{R}^n$ and $\Sigma \in \mathbb{R}^{n\times n}$ be symmetric and PSD (Lecture 8, Definition 13: $x'\Sigma x \geqslant 0$ for all x). z has the **multivariate normal distribution** $N_n(\mu, \Sigma)$, $z \sim N_n(\mu, \Sigma)$, if

$$M_z(t) = \exp\Big(t'\mu + \tfrac{1}{2} t'\Sigma t\Big) \qquad \text{for all } t \in \mathbb{R}^n$$

For $n = 1$: $N(\mu, \sigma^2)$, with the MGF $\exp(t\mu + \frac{1}{2}\sigma^2 t^2)$; $N(0, 1)$ is the **standard normal** distribution. u and v are **jointly normal** if $(u', v')'$ is multivariate normal.

Proposition 2. If $z \sim N_n(\mu, \Sigma)$, then $\mathsf{E}(z) = \mu$ and $\operatorname{var}(z) = \Sigma$.

Proof. The exponent $q(t) = t'\mu + \frac{1}{2} t'\Sigma t = \sum_k t_k \mu_k + \frac{1}{2} \sum_k \sum_l \sigma_{kl} t_k t_l$ ($\sigma_{kl} = \sigma_{lk}$) and the chain and product rules give

$$\frac{\partial q}{\partial t_i} = \mu_i + (\Sigma t)_i, \qquad \frac{\partial^2 q}{\partial t_j \partial t_i} = \sigma_{ij}, \qquad \frac{\partial M_z}{\partial t_i} = \Big(\mu_i + (\Sigma t)_i\Big) M_z(t)$$

$$\frac{\partial^2 M_z}{\partial t_j \partial t_i} = \Big(\sigma_{ij} + \Big(\mu_i + (\Sigma t)_i\Big)\Big(\mu_j + (\Sigma t)_j\Big)\Big) M_z(t)$$

At $t = 0$, $M_z(0) = 1$: $\mathsf{E}(z_i) = \mu_i$, $\mathsf{E}(z_i z_j) = \sigma_{ij} + \mu_i \mu_j$ (F1). So $\mathsf{E}(zz') = \Sigma + \mu\mu'$ and $\operatorname{var}(z) = \Sigma$ (Lecture 9, Theorem 4(d): $\mathsf{E}(uu') = \operatorname{var}(u) + \mu_u \mu_u'$). □

Theorem 3. Let $z \sim N_n(\mu, \Sigma)$, and $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$ constant. Then $Az + b \sim N_m(A\mu + b,\ A\Sigma A')$.

Proof.
$A\Sigma A' = (A')'\Sigma(A')$ is symmetric (Lecture 1, Theorem 8) and PSD (Lecture 8, Theorem 15(a): B PSD $\Rightarrow K'BK$ PSD, here $K = A'$); for $t \in \mathbb{R}^m$:

$$t'(Az + b) = t'b + (A't)'z \implies M_{Az+b}(t) = \exp(t'b)\, M_z(A't) \quad \text{(Lecture 1, Theorems 3(a), 8; F2)}$$

$$M_{Az+b}(t) = \exp\Big(t'b + (A't)'\mu + \tfrac{1}{2}(A't)'\Sigma(A't)\Big) \quad \text{(Definition 1)}$$

$$= \exp\Big(t'(A\mu + b) + \tfrac{1}{2} t'(A\Sigma A')t\Big) \quad \text{(Lecture 1, Theorems 3(a), 8)}$$

□

Corollary 4. Let $z \sim N_n(\mu, \Sigma)$.

(a) For $1 \leqslant i_1 < \cdots < i_k \leqslant n$: $(z_{i_1}, \dots, z_{i_k})' \sim N_k$, with the mean $(\mu_{i_1}, \dots, \mu_{i_k})'$ and the submatrix of Σ in the rows and columns $i_1, \dots, i_k$ as covariance matrix. In particular $z_i \sim N(\mu_i, \sigma_{ii})$; and for $z = (z_1', z_2')'$, $z_1 \in \mathbb{R}^p$, $z_2 \in \mathbb{R}^q$, $\Sigma_{11} \in \mathbb{R}^{p\times p}$,

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \implies z_1 \sim N_p(\mu_1, \Sigma_{11}), \quad z_2 \sim N_q(\mu_2, \Sigma_{22})$$

(b) $a'z \sim N(a'\mu,\ a'\Sigma a)$ for $a \in \mathbb{R}^n$.

Proof. (a) $T = [e_{i_1}\ \cdots\ e_{i_k}]$: $T'z = (z_{i_1}, \ldots, z_{i_k})'$, and $T'\Sigma T$ has the entries $e'_{i_r}\Sigma e_{i_s} = \sigma_{i_r i_s}$ (Lecture 1, Theorem 4(c),(d), Proposition 11(a)); Theorem 3 with $A = T'$, $b = 0$.

(b) Theorem 3 with $A = a'$, $b = 0$. □

Example 5. Let $z \sim N_4(\mu, \Sigma)$, with $\mu = (1, 0, 2, 2)'$ and the Σ of Lecture 9, Example 5,

$$\Sigma = \begin{pmatrix} 4 & 0 & 4 & 0 \\ 0 & 1 & 2 & 2 \\ 4 & 2 & 9 & 6 \\ 0 & 2 & 6 & 9 \end{pmatrix} = KK', \qquad K = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 \\ 0 & 2 & 2 & 1 \end{pmatrix}$$

($x'\Sigma x = (K'x)'(K'x) \geqslant 0$, a sum of squares (Lecture 1, Theorem 8): Σ is PSD.) Find the distributions of z_2, $(z_1, z_3)'$, $\mathbf{1}'z$ ($\mathbf{1} = (1, 1, 1, 1)'$), $e = z_3 - z_1 - 2z_2$, $f = z_4 + 2z_1 + 2z_2 - 2z_3$ and $(z_1 + z_2, z_1 - z_2)'$.

Solution.

Corollary 4; $e = a'z$, $f = b'z$ with $a = (-1, -2, 1, 0)'$, $b = (2, 2, -2, 1)'$; $a'\Sigma a = b'\Sigma b = 1$, $\mathbf{1}'\Sigma\mathbf{1} = 51$ (Lecture 9, Example 5):

$$z_2 \sim N(0, 1), \qquad \begin{pmatrix} z_1 \\ z_3 \end{pmatrix} \sim N_2\left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}\right) \qquad \text{(Corollary 4(a))}$$

$$\mathbf{1}'z \sim N(5, 51) \qquad \text{((b); } \mathbf{1}'\mu = 1 + 0 + 2 + 2 = 5)$$

$$e \sim N(1, 1) \qquad (a'\mu = -1 - 0 + 2 + 0 = 1)$$

$$f \sim N(0, 1) \qquad (b'\mu = 2 + 0 - 4 + 2 = 0)$$

$A = \left(\begin{smallmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{smallmatrix}\right)$, $Az = (z_1 + z_2, z_1 - z_2)'$; $A\Sigma A' = \operatorname{var}(Az)$ (Lecture 9, Theorem 3(b): $\operatorname{var}(Az + b) = A\operatorname{var}(z)A'$); $\operatorname{var}(z_1 + z_2) = \operatorname{var}(z_1 - z_2) = 5$, $\operatorname{cov}(z_1 + z_2, z_1 - z_2) = 3$ (Lecture 9, Example 5):

$$A\mu = \begin{pmatrix} 1 + 0 \\ 1 - 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad Az \sim N_2\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}\right) \qquad \text{(Theorem 3)}$$

2. Independence

F3. $u_1, \ldots, u_m$ **independent**: $\mathsf{P}(u_k \in A_k \text{ for all } k) = \prod_k \mathsf{P}(u_k \in A_k)$ (all A_k); then so are the $g_k(u_k)$.
F4. $u_1, \ldots, u_m$ independent $\Rightarrow \mathsf{E}\big(g_1(u_1) \cdots g_m(u_m)\big) = \mathsf{E}\, g_1(u_1) \cdots \mathsf{E}\, g_m(u_m)$; so $\operatorname{cov}(u_1, u_2) = 0$.
F5. MGFs finite near 0: $u_1, \ldots, u_m$ independent $\iff M_{(u'_1, \ldots, u'_m)'}(t_1, \ldots, t_m) = \prod_k M_{u_k}(t_k)$ near 0.

Theorem 6. (a) Let $z = (u', v')' \sim N_{p+q}(\mu, \Sigma)$, $u \in \mathbb{R}^p$, $v \in \mathbb{R}^q$, $\mu = (\mu'_u, \mu'_v)'$, $\Sigma = \left(\begin{smallmatrix} \Sigma_{uu} & \Sigma_{uv} \\ \Sigma_{vu} & \Sigma_{vv} \end{smallmatrix}\right)$ ($\Sigma_{uu} \in \mathbb{R}^{p \times p}$). Then u, v are independent (F3) $\iff \Sigma_{uv} = 0$ ($\iff \operatorname{cov}(u, v) = 0$).

(b) Let $D = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)$. Then $z \sim N_n(\mu, D) \iff z_1, \ldots, z_n$ are independent and $z_i \sim N(\mu_i, \sigma_i^2)$. In particular $z \sim N_n(\mu_0\mathbf{1}, \sigma^2 I_n)$ ($\mathbf{1} = \mathbf{1}_n$) $\iff z_1, \ldots, z_n$ are independent $N(\mu_0, \sigma^2)$ variables (a **random sample** from $N(\mu_0, \sigma^2)$).

(c) Let $z \sim N_n(\mu, \Sigma)$, and $A \in \mathbb{R}^{p \times n}$, $B \in \mathbb{R}^{q \times n}$ constant. Then Az, Bz are independent $\iff A\Sigma B' = 0$.

Proof. (a) $\Rightarrow$: $\Sigma = \operatorname{var}(z)$ (Proposition 2) has the block $\Sigma_{uv} = \operatorname{cov}(u, v)$ (Lecture 9, Definition 1: $\sigma_{ij} = \operatorname{cov}(z_i, z_j)$; Definition 2: $\operatorname{cov}(u, v)$ has the entries $\operatorname{cov}(u_i, v_j)$), which is 0 if u, v are independent (F4).

(a) $\Leftarrow$: $\Sigma_{uv} = 0$, so $\Sigma_{vu} = \Sigma_{uv}' = 0$; for $s \in \mathbb{R}^p$, $t \in \mathbb{R}^q$, by blocks (Lecture 5, Theorem 2(c): blocks multiply block by block):

$$(s', t')\mu = s'\mu_u + t'\mu_v, \qquad (s', t')\,\Sigma \begin{pmatrix} s \\ t \end{pmatrix} = s'\Sigma_{uu}s + t'\Sigma_{vv}t$$

$$M_z(s,t) = \exp\Big(s'\mu_u + \tfrac{1}{2}s'\Sigma_{uu}s\Big)\exp\Big(t'\mu_v + \tfrac{1}{2}t'\Sigma_{vv}t\Big) = M_u(s)M_v(t) \qquad \text{(Corollary 4(a))}$$

For all s, t: u, v are independent (F5).

(b) $t'Dt = \sum_i \sigma_i^2 t_i^2$ (Lecture 1, Proposition 13(a), Theorem 8), so D is PSD; $\Rightarrow$ uses $z_i \sim N(\mu_i, \sigma_i^2)$ (Corollary 4(a)), $\Leftarrow$ uses F4; for all $t \in \mathbb{R}^n$:

$$\Rightarrow \quad M_z(t) = \exp\Big(t'\mu + \tfrac{1}{2}t'Dt\Big) = \textstyle\prod_i \exp\Big(t_i\mu_i + \tfrac{1}{2}\sigma_i^2 t_i^2\Big) = \prod_i M_{z_i}(t_i)$$

$$\Leftarrow \quad M_z(t) = \mathsf{E}\textstyle\prod_i e^{t_i z_i} = \prod_i \mathsf{E}\, e^{t_i z_i} = \prod_i \exp\Big(t_i\mu_i + \tfrac{1}{2}\sigma_i^2 t_i^2\Big) = \exp\Big(t'\mu + \tfrac{1}{2}t'Dt\Big)$$

$\Rightarrow$: $z_1, \ldots, z_n$ are independent (F5). $\Leftarrow$: $z \sim N_n(\mu, D)$ (Definition 1).

(c) $G = \left(\begin{smallmatrix} A \\ B \end{smallmatrix}\right) \in \mathbb{R}^{(p+q)\times n}$, $Gz = \Big((Az)', (Bz)'\Big)'$; blocks (Lecture 5, Theorem 2(b),(c): transposed and multiplied block by block):

$$Gz \sim N_{p+q}(G\mu, G\Sigma G'), \qquad G\Sigma G' = \begin{pmatrix} A\Sigma A' & A\Sigma B' \\ B\Sigma A' & B\Sigma B' \end{pmatrix} \qquad \text{(Theorem 3)}$$

By (a) with $u = Az$, $v = Bz$: independent $\iff A\Sigma B' = 0$.

□

Remark 7. Joint normality is needed in Theorem 6(a). Let u take the values -1, 0, 1, each with probability $\frac{1}{3}$, and $v = u^2$. By F2, $\mathsf{E}(u) = \mathsf{E}(u^3) = 0$, $\mathsf{E}(u^2) = \frac{2}{3}$, and $\operatorname{cov}(u, v) = \mathsf{E}(u^3) - \mathsf{E}(u)\,\mathsf{E}(u^2) = 0$ (Lecture 9, Definition 2: $\operatorname{cov}(u, v) = \mathsf{E}\Big((u - \mu_u)(v - \mu_v)'\Big)$; F2); but $\mathsf{P}(u = 0,\ v = 1) = 0 \neq \frac{1}{3} \cdot \frac{2}{3} = \mathsf{P}(u = 0)\,\mathsf{P}(v = 1)$ (F3).

Example 8. For z of Example 5: are z_1 and $(z_2, z_4)'$ independent? e and $(z_1, z_2)'$? f and $(z_1, z_2, z_3)'$? $z_1 + z_2$ and $z_1 - z_2$?

Solution.

Corollary 4(a) with $i_1, i_2, i_3 = 1, 2, 4$:

$$\begin{pmatrix} z_1 \\ z_2 \\ z_4 \end{pmatrix} \sim N_3\left(\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 9 \end{pmatrix}\right)$$

$\sigma_{12} = \sigma_{14} = 0$: z_1 and $(z_2, z_4)'$ are independent (Theorem 6(a), $p = 1$, $q = 2$).

Theorem 6(c) with $B = a'$, $B = b'$; $\Sigma a = (0, 0, 1, 2)'$, $\Sigma b = (0, 0, 0, 1)'$ (Lecture 9, Example 5):

$$[I_2 \ \ 0]\,\Sigma a = (0, 0)' \implies e \text{ and } (z_1, z_2)' \text{ independent}, \qquad z_3 = z_1 + 2z_2 + e$$

$$[I_3 \ \ 0]\,\Sigma b = (0, 0, 0)' \implies f \text{ and } (z_1, z_2, z_3)' \text{ independent}, \qquad z_4 = -2z_1 - 2z_2 + 2z_3 + f$$

$\operatorname{cov}(z_1 + z_2, z_1 - z_2) = 3 \neq 0$: $z_1 + z_2$ and $z_1 - z_2$ are not independent (F4).

3. Conditional distributions

Theorem 9. Let $z \sim N_n(\mu, \Sigma)$ be partitioned as in Corollary 4(a), with Σ_{11} PD (Lecture 8, Definition 13: $x'\Sigma_{11}x > 0$ for $x \neq 0$), and $\Sigma_{22\cdot1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$. Then $\Sigma_{22\cdot1}$ is PSD, and

$$w = z_2 - \mu_2 - \Sigma_{21}\Sigma_{11}^{-1}(z_1 - \mu_1) \sim N_q(0, \Sigma_{22\cdot1}), \qquad w \text{ and } z_1 \text{ independent}$$

So $z_2 = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(z_1 - \mu_1) + w$. For $a \in \mathbb{R}^p$, $N_q\Big(\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(a - \mu_1),\ \Sigma_{22\cdot1}\Big)$ is the **conditional distribution of z_2 given $z_1 = a$.**

Proof.

Σ_{11}^{-1} exists (Lecture 8, Theorem 14(c): PD $\Rightarrow$ invertible) and is symmetric (Lecture 2, Corollary 12); $L = \left(\begin{smallmatrix} I_p & 0 \\ -\Sigma_{21}\Sigma_{11}^{-1} & I_q \end{smallmatrix}\right)$, $L' = \left(\begin{smallmatrix} I_p & -\Sigma_{11}^{-1}\Sigma_{12} \\ 0 & I_q \end{smallmatrix}\right)$ ($\Sigma_{21}' = \Sigma_{12}$); blocks (Lecture 5, Theorem 2(b),(c): transposed and multiplied block by block):

$$L(z - \mu) = \begin{pmatrix} z_1 - \mu_1 \\ -\Sigma_{21}\Sigma_{11}^{-1}(z_1 - \mu_1) + z_2 - \mu_2 \end{pmatrix} = \begin{pmatrix} z_1 - \mu_1 \\ w \end{pmatrix}$$

$$L\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ 0 & \Sigma_{22\cdot1} \end{pmatrix}, \qquad L\Sigma L' = (L\Sigma)L' = \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22\cdot1} \end{pmatrix}$$

(Lecture 5, Theorem 3: for $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, A invertible, $S = D - CA^{-1}B$, $\left(\begin{smallmatrix} I & 0 \\ -CA^{-1} & I \end{smallmatrix}\right) M = \left(\begin{smallmatrix} A & B \\ 0 & S \end{smallmatrix}\right)$, with $M = \Sigma$, gives $L\Sigma$; $M \left(\begin{smallmatrix} I & -A^{-1}B \\ 0 & I \end{smallmatrix}\right) = \left(\begin{smallmatrix} A & 0 \\ C & S \end{smallmatrix}\right)$, with $M = L\Sigma$, $C = 0$, gives $L\Sigma L'$.)

Theorem 3 with $A = L$, $b = -L\mu$:

$$L(z - \mu) \sim N_n(0, L\Sigma L') \implies w \sim N_q(0, \Sigma_{22\cdot1}) \qquad \text{(Corollary 4(a))}$$

$$\Sigma_{22\cdot1} = \text{var}(w) \text{ is PSD} \qquad \text{(Proposition 2; Lecture 9, Theorem 3(c))}$$

The off-diagonal blocks of $L\Sigma L'$ are 0: $z_1 - \mu_1$ and w are independent (Theorem 6(a)), and so are $z_1 = (z_1 - \mu_1) + \mu_1$ and w (F3).

□

The conditional mean is an affine function of a, and $\Sigma_{22\cdot1}$ does not depend on a. With $K = \Sigma_{11}^{-1}\Sigma_{12}$, $K'\Sigma_{11}K = \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{11}\Sigma_{11}^{-1}\Sigma_{12} = \Sigma_{22} - \Sigma_{22\cdot1}$ is PSD (Lecture 8, Theorem 15(a): B PSD $\Rightarrow$ $K'BK$ PSD); so $\text{var}(c'w) = c'\Sigma_{22\cdot1}c \leqslant c'\Sigma_{22}c = \text{var}(c'z_2)$ for $c \in \mathbb{R}^q$ (Lecture 9, Theorem 3(b)): knowing z_1 can only reduce the variance.

Example 10. For z of Example 5, find the conditional distribution of $(z_3, z_4)'$ given $z_1 = a_1$, $z_2 = a_2$.

Solution.

Theorem 9 with $(z_1, z_2)'$, $(z_3, z_4)'$ as its z_1, z_2, $p = q = 2$; the blocks of Σ (Example 5); $x'\Sigma_{11}x = 4x_1^2 + x_2^2 > 0$ for $x \neq 0$: Σ_{11} is PD:

$$\Sigma_{11} = \text{diag}(4, 1), \qquad \Sigma_{11}^{-1} = \text{diag}(\tfrac{1}{4}, 1), \qquad \Sigma_{21} = \begin{pmatrix} 4 & 2 \\ 0 & 2 \end{pmatrix}, \qquad \Sigma_{21}\Sigma_{11}^{-1} = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}$$

$$\Sigma_{22\cdot1} = \begin{pmatrix} 9 & 6 \\ 6 & 9 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 4 & 0 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 9 & 6 \\ 6 & 9 \end{pmatrix} - \begin{pmatrix} 8 & 4 \\ 4 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$$

The conditional mean (Theorem 9), $\mu_1 = (1, 0)'$, $\mu_2 = (2, 2)'$:

$$\begin{pmatrix} 2 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} a_1 - 1 \\ a_2 \end{pmatrix} = \begin{pmatrix} a_1 + 2a_2 + 1 \\ 2a_2 + 2 \end{pmatrix}, \qquad N_2\left(\begin{pmatrix} a_1 + 2a_2 + 1 \\ 2a_2 + 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}\right)$$

The first entry agrees with Example 8: $z_3 = z_1 + 2z_2 + e$, $e \sim N(1, 1)$ independent of $(z_1, z_2)'$. Knowing z_1, z_2 reduces the variances of z_3, z_4 from 9, 9 to 1, 5.

4. The t and F distributions

F6. χ^2_r (**chi-square**), t_ν, $F_{a,b}$ ($r, \nu, a, b \geqslant 1$): the distributions of $w_1^2 + \cdots + w_r^2$, $Z/\sqrt{V/\nu}$, $(U/a)/(V/b)$ for independent $w_i \sim N(0,1)$; independent $Z \sim N(0,1)$, $V \sim \chi^2_\nu$; independent $U \sim \chi^2_a$, $V \sim \chi^2_b$.
F7. y, z have the **same distribution** if $\mathsf{P}(y \in A) = \mathsf{P}(z \in A)$ (all A); e.g. if $M_y = M_z$ finite near 0.
F8. For u as in F6, $G(x) = \mathsf{P}(u \leqslant x)$ is continuous ($\mathsf{P}(u = c) = 0$), strictly increasing on $\mathbb{R}$ (t_ν) or on $[0, \infty)$, $G(0) = 0$; the τ **quantile** $\chi^2_{r,\tau}$, $t_{\nu,\tau}$, $F_{a,b,\tau}$ ($0 < \tau < 1$) is the c with $G(c) = \tau$.
F9. $\mathsf{P}(z \in A \cup B) = \mathsf{P}(z \in A) + \mathsf{P}(z \in B)$ for disjoint A, B; $\mathsf{P}(z \in \mathbb{R}^n) = 1$.

Theorem 11. Let $T \sim t_\nu$ and $\alpha \in (0,1)$.

(a) $-T \sim t_\nu$; $\mathsf{P}(T < -c) = \mathsf{P}(T > c)$ for $c \in \mathbb{R}$; and

$$\mathsf{P}(T < -t_{\nu,1-\alpha}) = \alpha, \qquad \mathsf{P}\Big(-t_{\nu,1-\alpha/2} \leqslant T \leqslant t_{\nu,1-\alpha/2}\Big) = 1 - \alpha$$

(b) $T^2 \sim F_{1,\nu}$; $t_{\nu,1-\alpha/2} > 0$ and $t^2_{\nu,1-\alpha/2} = F_{1,\nu,1-\alpha}$.

Proof.

(a) $T = Z/\sqrt{V/\nu}$, $Z \sim N(0,1)$, $V \sim \chi^2_\nu$ independent (F6); $-Z \sim N(0,1)$ (Theorem 3, $A = -1$), independent of V (F3); $k = t_{\nu,1-\alpha/2}$:

$$\begin{aligned}
-T = (-Z)/\sqrt{V/\nu} \sim t_\nu & \qquad \text{(F6)} \\
\mathsf{P}(T < -c) = \mathsf{P}(-T > c) = \mathsf{P}(T > c) & \qquad \text{(F7)} \\
\mathsf{P}(T < -t_{\nu,1-\alpha}) = \mathsf{P}(T > t_{\nu,1-\alpha}) = \alpha & \qquad \text{(F8, F9)} \\
\mathsf{P}(-k \leqslant T \leqslant k) = 1 - \mathsf{P}(T < -k) - \mathsf{P}(T > k) = 1 - \tfrac{\alpha}{2} - \tfrac{\alpha}{2} & \qquad \text{(F9; line 2 with } c = k\text{)}
\end{aligned}$$

(b) $Z^2 \sim \chi^2_1$ (F6), independent of V (F3); $k = t_{\nu,1-\alpha/2}$:

$$\begin{aligned}
T^2 = (Z^2/1)/(V/\nu) \sim F_{1,\nu} & \qquad \text{(F6)} \\
\mathsf{P}(T < 0) = \mathsf{P}(T > 0), \quad \mathsf{P}(T = 0) = 0 \implies \mathsf{P}(T \leqslant 0) = \tfrac{1}{2} & \qquad \text{((a), } c = 0\text{; F8, F9)} \\
\mathsf{P}(T \leqslant 0) = \tfrac{1}{2} < 1 - \tfrac{\alpha}{2} = \mathsf{P}(T \leqslant k) \implies k > 0 & \qquad \text{(F8: increasing)} \\
\mathsf{P}(T^2 \leqslant k^2) = \mathsf{P}(-k \leqslant T \leqslant k) = 1 - \alpha \implies k^2 = F_{1,\nu,1-\alpha} & \qquad \text{((a); F8)}
\end{aligned}$$

□

5. Standardization, the density and quadratic forms

Proposition 12. Let Σ be PD, $\Sigma = Q\Lambda Q'$ with Q orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ (Lecture 8, Theorem 9(a),(b)), so $\lambda_i > 0$ (Lecture 8, Theorem 14(a): PD $\iff$ every $\lambda_i > 0$); $\Lambda^{\pm 1/2} = \operatorname{diag}(\lambda_1^{\pm 1/2}, \dots, \lambda_n^{\pm 1/2})$, $\Sigma^{1/2} = Q\Lambda^{1/2}Q'$, $\Sigma^{-1/2} = Q\Lambda^{-1/2}Q'$. Then $\Sigma^{1/2}$, $\Sigma^{-1/2}$ are symmetric, and

$$\Sigma^{1/2}\Sigma^{1/2} = \Sigma, \qquad \Sigma^{1/2}\Sigma^{-1/2} = I, \qquad \Sigma^{-1/2}\Sigma\Sigma^{-1/2} = I, \qquad \Sigma^{-1/2}\Sigma^{-1/2} = \Sigma^{-1}$$

Proof.

$Q'Q = I = QQ'$ (Lecture 6, Theorem 9(b): $Q'Q = I_n \Rightarrow QQ' = I_n$); products of diagonal matrices (Lecture 1, Proposition 13(b)); $BA = I \Rightarrow B = A^{-1}$ (Lecture 2, Theorem 15):

$$\begin{aligned}
(Q\Lambda^{\pm 1/2}Q')' &= Q(\Lambda^{\pm 1/2})'Q' = Q\Lambda^{\pm 1/2}Q' && \text{(Lecture 1, Theorem 8)}\\
\Sigma^{1/2}\Sigma^{1/2} &= Q\Lambda^{1/2}Q'Q\Lambda^{1/2}Q' = Q\Lambda Q' = \Sigma\\
\Sigma^{1/2}\Sigma^{-1/2} &= Q\Lambda^{1/2}\Lambda^{-1/2}Q' = QQ' = I\\
\Sigma^{-1/2}\Sigma\Sigma^{-1/2} &= Q\Lambda^{-1/2}\Lambda\Lambda^{-1/2}Q' = QQ' = I\\
(\Sigma^{-1/2}\Sigma^{-1/2})\Sigma &= Q\Lambda^{-1}Q'Q\Lambda Q' = QQ' = I \implies \Sigma^{-1/2}\Sigma^{-1/2} = \Sigma^{-1}
\end{aligned}$$

□

Theorem 13. Let $z \sim N_n(\mu, \Sigma)$, Σ PD, and $w = \Sigma^{-1/2}(z - \mu)$. Then $w \sim N_n(0, I_n)$: the w_i are independent and $N(0,1)$; $z = \mu + \Sigma^{1/2}w$; and $(z-\mu)'\Sigma^{-1}(z-\mu) = w'w \sim \chi^2_n$.

Proof.

Theorem 3 with $A = \Sigma^{-1/2}$, $b = -\Sigma^{-1/2}\mu$; $\Sigma^{-1/2}$ is symmetric (Proposition 12):

$$\begin{aligned}
w \sim N_n\Big(\Sigma^{-1/2}\mu - \Sigma^{-1/2}\mu,\ \Sigma^{-1/2}\Sigma\Sigma^{-1/2}\Big) &= N_n(0, I_n) && \text{(Proposition 12)}\\
\implies w_1, \dots, w_n \text{ independent, } N(0,1) && && \text{(Theorem 6(b))}\\
\Sigma^{1/2}w = \Sigma^{1/2}\Sigma^{-1/2}(z-\mu) &= z - \mu\\
w'w = (z-\mu)'\Sigma^{-1/2}\Sigma^{-1/2}(z-\mu) &= (z-\mu)'\Sigma^{-1}(z-\mu) \sim \chi^2_n && \text{(Lecture 1, Theorem 8; F6)}
\end{aligned}$$

□

So $\mathsf{P}\Big((z-\mu)'\Sigma^{-1}(z-\mu) \leqslant \chi^2_{n,1-\alpha}\Big) = 1 - \alpha$ (F8).

F10. $\int_{\mathbb{R}^n} g(x)\,\mathrm{d}x = |\det B| \int_{\mathbb{R}^n} g(c + Bu)\,\mathrm{d}u$ (B invertible); $\int_{\mathbb{R}^n} e^{-u'u/2}\,\mathrm{d}u = \Big(\int_{\mathbb{R}} e^{-s^2/2}\,\mathrm{d}s\Big)^n = (2\pi)^{n/2}$.
F11. z has **density** $f \geqslant 0$ if $\mathsf{P}(z \in A) = \int_A f$ (all A); $\mathsf{E}\,g(z) = \int gf$; any $f \geqslant 0$ with $\int f = 1$ is one.

Theorem 14. Let $\mu \in \mathbb{R}^n$, $\Sigma \in \mathbb{R}^{n\times n}$ PD, and

$$f(x) = \frac{1}{(2\pi)^{n/2}(\det\Sigma)^{1/2}} \exp\Big(-\frac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)\Big) \qquad (x \in \mathbb{R}^n)$$

(a) $f(x) > 0$ for every x, and $\int_{\mathbb{R}^n} f(x)\,\mathrm{d}x = 1$.

(b) A random vector with the density f has the distribution $N_n(\mu, \Sigma)$; so $N_n(\mu, \Sigma)$ exists.

(c) Every $z \sim N_n(\mu, \Sigma)$ has the density f.

For $n = 1$, $\sigma^2 > 0$: $N(\mu, \sigma^2)$ has the density $(2\pi\sigma^2)^{-1/2}\exp\Big(-(x-\mu)^2/(2\sigma^2)\Big)$.

Proof. (a) $\det\Sigma > 0$ (Lecture 8, Theorem 14(c): PD $\Rightarrow$ det > 0): f is defined, $f(x) > 0$. With Q, Λ of Proposition 12, $Q'\Sigma^{1/2}Q = \Lambda^{1/2}$ and $\lambda_i^{1/2} > 0$, so $\Sigma^{1/2}$ is PD (Lecture 8, Theorem 14(a)), $\det\Sigma^{1/2} > 0$ and $\Sigma^{1/2}$ is invertible (Lecture 8, Theorem 14(c)); $(\det\Sigma^{1/2})^2 = \det\Sigma$ (Lecture 5, Theorem 12(b): $\det(AB) = \det A \det B$), so $\det\Sigma^{1/2} = (\det\Sigma)^{1/2}$. For $x = \mu + \Sigma^{1/2}u$: $(x-\mu)'\Sigma^{-1}(x-\mu) = u'(\Sigma^{1/2}\Sigma^{-1/2})(\Sigma^{1/2}\Sigma^{-1/2})'u = u'u$ (Proposition 12; Lecture 1, Theorem 8).

F10 with $g = f$, $c = \mu$, $B = \Sigma^{1/2}$:

$$\int_{\mathbb{R}^n} f(x)\,\mathrm{d}x = \frac{(\det\Sigma)^{1/2}}{(2\pi)^{n/2}(\det\Sigma)^{1/2}} \int_{\mathbb{R}^n} e^{-u'u/2}\,\mathrm{d}u = \frac{(2\pi)^{n/2}}{(2\pi)^{n/2}} = 1$$

(b) For $t, x \in \mathbb{R}^n$, expanding with $\Sigma' = \Sigma$, $\Sigma^{-1}\Sigma = I$ (Lecture 1, Theorems 3(a), 8): $(x-\mu-\Sigma t)'\Sigma^{-1}(x-\mu-\Sigma t) = (x-\mu)'\Sigma^{-1}(x-\mu) - 2t'(x-\mu) + t'\Sigma t$. So $e^{t'x}f(x) = \exp(t'\mu + \frac{1}{2}t'\Sigma t)f_t(x)$, where f_t is f with $\mu + \Sigma t$ for μ. If y has the density f: $M_y(t) = \int_{\mathbb{R}^n} e^{t'x}f(x)\,\mathrm{d}x = \exp(t'\mu + \frac{1}{2}t'\Sigma t)\int_{\mathbb{R}^n} f_t(x)\,\mathrm{d}x = \exp(t'\mu + \frac{1}{2}t'\Sigma t)$ (F11; (a) for $\mu + \Sigma t$): $y \sim N_n(\mu, \Sigma)$. By (a) and F11, such a y exists.

(c) For y as in (b), $M_z = M_y$ (Definition 1): z, y have the same distribution (F7), and $\mathsf{P}(z \in A) = \mathsf{P}(y \in A) = \int_A f(x)\,\mathrm{d}x$ for every A. For $n = 1$: $\det(\sigma^2) = \sigma^2$, $(\sigma^2)^{-1} = 1/\sigma^2$. □

Theorem 15. Let $z \sim N_n(\mu, \sigma^2 I_n)$ ($\sigma > 0$), and $A \in \mathbb{R}^{n\times n}$ symmetric, $A^2 = A$, $\operatorname{rank} A = r \geqslant 1$, $A\mu = 0$. Then $z'Az/\sigma^2 \sim \chi^2_r$.

Proof.

$q_1, \ldots, q_n$ orthonormal, $Aq_i = \lambda_i q_i$ (Lecture 8, Theorem 9(a),(d): such q_i exist, and then $A = \sum_i \lambda_i q_i q_i'$), $\lambda_i \in \{0, 1\}$ (Lecture 8, Theorem 11: $A^2 = A \iff$ every eigenvalue is 0 or 1), $\#\{i : \lambda_i = 1\} = \operatorname{rank} A = r$ (Lecture 8, Theorem 9(b): $\operatorname{rank} A = \#\{i : \lambda_i \neq 0\}$); number the q_i so that $\lambda_1 = \cdots = \lambda_r = 1$; $Q_1 = [q_1 \ \cdots \ q_r]$, $Q_1'Q_1 = I_r$ (Lecture 6, Definition 8):

$$\begin{aligned}
A &= q_1q_1' + \cdots + q_rq_r' = Q_1Q_1' && \text{(Lecture 1, Theorem 4(d))}\\
Q_1'\mu &= Q_1'Q_1Q_1'\mu = Q_1'A\mu = 0 &&\\
u &= \tfrac{1}{\sigma}Q_1'z \sim N_r\Big(0, \tfrac{1}{\sigma^2}Q_1'(\sigma^2 I_n)Q_1\Big) = N_r(0, I_r) && \text{(Theorem 3)}\\
z'Az/\sigma^2 &= (Q_1'z)'(Q_1'z)/\sigma^2 = u'u = u_1^2 + \cdots + u_r^2 \sim \chi^2_r && \text{(Theorem 6(b); F6)}
\end{aligned}$$

□

Theorem 16. Let $z \sim N_n(\mu, \sigma^2 I_n)$, and $A, B \in \mathbb{R}^{n\times n}$ symmetric with $A^2 = A$, $B^2 = B$.

(a) If $AB = 0$, then $z'Az$ and $z'Bz$ are independent.

(b) If $K \in \mathbb{R}^{m\times n}$ and $KA = 0$, then Kz and $z'Az$ are independent.

Proof.

(a) $A'A = A^2 = A$: $z'Az = z'A'Az = (Az)'(Az)$, a function of Az; likewise $z'Bz = (Bz)'(Bz)$; Theorem 6(c), F3:

$$A(\sigma^2 I_n)B' = \sigma^2 AB = 0 \implies Az,\ Bz \text{ independent} \implies z'Az,\ z'Bz \text{ independent}$$

(b) $z'Az = (Az)'(Az)$ as in (a); Theorem 6(c), F3:

$$K(\sigma^2 I_n)A' = \sigma^2 KA = 0 \implies Kz,\ Az \text{ independent} \implies Kz,\ z'Az \text{ independent}$$

□

6. One normal sample

Theorem 17. Let $n \geqslant 2$, $x_1, \ldots, x_n$ independent $N(\mu, \sigma^2)$ variables ($\sigma > 0$), $\bar{x} = \frac{1}{n}\sum_i x_i$, $s^2 = \frac{1}{n-1}\sum_i (x_i - \bar{x})^2$ and $s = \sqrt{s^2}$.

(a) $\bar{x} \sim N(\mu, \sigma^2/n)$.

(b) $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$.

(c) $\bar{x}$ and s^2 are independent.

(d) $\sqrt{n}(\bar{x} - \mu)/s \sim t_{n-1}$.

Proof. $x = (x_1, \ldots, x_n)' \sim N_n(\mu\mathbf{1}, \sigma^2 I_n)$ (Theorem 6(b)), and with $C_n = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}'$ (Lecture 1, Definition 14), $(n-1)s^2 = x'C_n x$ (Lecture 1, Example 16).

(a) Corollary 4(b) with $a = \frac{1}{n}\mathbf{1}$: $a'x = \bar{x}$, $a'(\mu\mathbf{1}) = \mu$, $a'(\sigma^2 I_n)a = \sigma^2/n$.

(b) $C_n' = C_n = C_n^2$, $C_n\mathbf{1} = 0$ (Lecture 1, Proposition 15), $\operatorname{rank} C_n = n - 1 \geqslant 1$ (Lecture 8, Example 5):

$$\frac{(n-1)s^2}{\sigma^2} = \frac{x'C_n x}{\sigma^2} \sim \chi^2_{n-1} \qquad \text{(Theorem 15, } A = C_n\text{)}$$

(c) $K = \frac{1}{n}\mathbf{1}'$: $Kx = \bar{x}$, $KC_n = \frac{1}{n}(C_n\mathbf{1})' = 0$ (Lecture 1, Theorem 8, Proposition 15):

$$\bar{x} \text{ and } x'C_n x \text{ independent} \implies \bar{x} \text{ and } s^2 = \tfrac{1}{n-1}x'C_n x \text{ independent} \qquad \text{(Theorem 16(b); F3)}$$

(d) $Z = \sqrt{n}(\bar{x} - \mu)/\sigma \sim N(0,1)$ ((a); Theorem 3), $V = (n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$ ((b)), independent ((c); F3); $\mathsf{P}(V = 0) = 0$ (F8), so $s > 0$ with probability 1:

$$\frac{\sqrt{n}(\bar{x} - \mu)}{s} = \frac{\sqrt{n}(\bar{x} - \mu)/\sigma}{\sqrt{(n-1)s^2/\big((n-1)\sigma^2\big)}} = \frac{Z}{\sqrt{V/(n-1)}} \sim t_{n-1} \qquad \text{(F6)}$$

□

Definition 18. Let observations be modeled as random variables whose distribution depends on parameters, and $\alpha \in (0,1)$. An interval $[L, U]$ whose endpoints $L \leqslant U$ are computed from the observations is a **confidence interval** with level $1-\alpha$ for a parameter $\theta \in \mathbb{R}$ if $\mathsf{P}(L \leqslant \theta \leqslant U) = 1 - \alpha$ for all values of the parameters.

Example 19. In Theorem 17, let $\alpha \in (0,1)$, $T = \sqrt{n}(\bar{x} - \mu)/s$, $V = (n-1)s^2/\sigma^2$. Find confidence intervals with level $1-\alpha$ for μ and for σ^2; compute them for five measurements $2, 4, 5, 6, 8$, modelled as independent $N(\mu, \sigma^2)$ variables; when does 3 lie in the interval for μ?

Solution.

$c = t_{n-1,1-\alpha/2} > 0$, $T \sim t_{n-1}$ (Theorems 11(b), 17(d)); $s > 0$:

$$\mathsf{P}(-c \leqslant T \leqslant c) = 1 - \alpha \qquad \text{(Theorem 11(a); all } \mu, \sigma\text{)}$$

$$-c \leqslant \frac{\sqrt{n}(\bar{x} - \mu)}{s} \leqslant c \iff \bar{x} - \frac{cs}{\sqrt{n}} \leqslant \mu \leqslant \bar{x} + \frac{cs}{\sqrt{n}}$$

So $[\bar{x} - cs/\sqrt{n},\ \bar{x} + cs/\sqrt{n}]$ is a confidence interval for μ with level $1-\alpha$ (Definition 18).

$c_1 = \chi^2_{n-1,\alpha/2}$, $c_2 = \chi^2_{n-1,1-\alpha/2}$, $V \sim \chi^2_{n-1}$ (Theorem 17(b)); F8, F9:

$$\mathsf{P}(c_1 \leqslant V \leqslant c_2) = \mathsf{P}(V \leqslant c_2) - \mathsf{P}(V < c_1) = 1 - \tfrac{\alpha}{2} - \tfrac{\alpha}{2} \qquad (\mathsf{P}(V = c_1) = 0)$$

$$\mathsf{P}(V \leqslant 0) = 0 < \tfrac{\alpha}{2} < 1 - \tfrac{\alpha}{2} \implies 0 < c_1 < c_2$$

$$c_1 \leqslant \frac{(n-1)s^2}{\sigma^2} \leqslant c_2 \iff \frac{(n-1)s^2}{c_2} \leqslant \sigma^2 \leqslant \frac{(n-1)s^2}{c_1}$$

So $[(n-1)s^2/c_2,\ (n-1)s^2/c_1]$ is a confidence interval for σ^2 with level $1-\alpha$.

The data, $n = 5$:

$$\bar{x} = \tfrac{25}{5} = 5, \qquad x_i - \bar{x}: \ -3, -1, 0, 1, 3, \qquad s^2 = \tfrac{9+1+0+1+9}{4} = 5$$
$$\tfrac{s}{\sqrt{5}} = 1, \qquad (n-1)s^2 = 20$$
$$\mu: \ \Big[5 - t_{4,1-\alpha/2}, \ 5 + t_{4,1-\alpha/2}\Big], \qquad \sigma^2: \ \Big[20/\chi^2_{4,1-\alpha/2}, \ 20/\chi^2_{4,\alpha/2}\Big]$$
$$\mu = 3: \ \frac{\sqrt{5}(\bar{x} - 3)}{s} = \frac{\sqrt{5} \cdot 2}{\sqrt{5}} = 2, \qquad 3 \in \Big[5 - t_{4,1-\alpha/2}, \ 5 + t_{4,1-\alpha/2}\Big] \iff t_{4,1-\alpha/2} \geqslant 2$$

Practice. The Exercises and Problems of the textbook, Chapters 10 and 11: normal vectors, linear transformations and marginal distributions, independence, conditional distributions, the density, the chi-square, t and F distributions, quadratic forms in normal vectors, and one normal sample.